

Ernest Wong

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EDUCATION

Simon Fraser University, Burnaby, BC April 2027
Bachelor of Science in Computer Science

University of Glasgow, Scotland, UK August 2024
Field Studies in Cybercrime

British Columbia Institute of Technology, Vancouver, BC December 2021
Diploma in Cloud Computing Systems, Completed with Distinction

TECHNICAL EXPERIENCE

Elections Canada | Government of Canada May 2026 - Present
Data Analyst Co-op

- Translated complex Power BI transformation logic into reusable Denodo VQL and SQL views, supporting automated reporting and data analysis.
- Built data-validation workflows to compare records across Denodo, People Insights, and source systems, identifying missing or inconsistent results.
- Collaborated with analysts and senior stakeholders to translate operational requirements into maintainable data products, reporting rules, and automation tools.

ISED | Government of Canada January 2025 - Present
Software Engineer Intern

- Led production deployments for the \$100M+ revenue-generating patent submission portal, coordinating AWS Cloud Ops, Salesforce, and Java teams to ship releases cleanly within maintenance windows.
- Built Java drift detection tools using GraphQL APIs to monitor configuration consistency across 8+ environments with Jenkins automation for daily health reports and send mismatch alerts.
- Developed Python/Selenium release automation tools to standardize CI/CD setup across Bitbucket, Bamboo, Nexus, and IBM WebSphere for 50+ applications, reducing manual release workflows.
- Resolved UI/UX and API defects in Salesforce Lightning components affecting bilingual submissions, improving reliability for high-volume online patent filings.
- Co-designed a Duolingo-inspired equity, diversity, and inclusion (EDI) training module in under 6 hours, earning finalist honors for “Most Impactful” and “Best Use of Google Gemini” at Canada’s largest AI hackathon and featured in the ISED newsletter.

UBC Orbit | SFU SAT Engineering Design Team September 2023 - Present
Mission Operations Developer | Embedded Systems Lead | Management

- Built the ALEASAT Spacecraft Dashboard (AWS, OpenShift API, React) to visualize telemetry and run command workflows, supporting a European Space Agency mission targeting a 2028 SpaceX launch.
- Migrated outreach websites from Squarespace to GitHub Pages and React, eliminating hosting costs.
- Spearheaded the full design lifecycle (Preliminary Design Review, Critical Design Review, Flight Readiness Review) for a high-altitude balloon reaching heights of 30km.
- Developed python software integrating 10+ sensors (GPS, Cameras, Transmitters) with a Raspberry Pi.

École Montgomery Middle School

September 2024 – December 2024

After-school Computer Instructor

- Designed and delivered coding lessons for a class of 20 students using Minecraft Education, block-based programming, and Python.
- Introduced web development through CodePen and GitHub Pages, enabling students to design and publish their own websites.
- Adapted technical concepts into accessible, hands-on activities that encouraged experimentation, creativity, and independent problem-solving.

Simon Fraser University | Faculty of Applied Sciences

May 2024 - August 2024

Research Assistant in AI Cybersecurity

- Developed a machine learning tool using Python to remove personal information from AI models achieving F1 greater than 80% on evaluation datasets.
- Identified potential security risks in AI models, such as unauthorized access to sensitive data, using Tensor Trust to demonstrate vulnerabilities.

LEADERSHIP & TEACHING

Simon Fraser University | Faculty of Applied Sciences

January 2024 - Present

Student Ambassador and Computing Science Peer Tutor

- Delivered engaging presentations at university recruitment events, reaching 100+ prospective students, inspiring two individuals to pursue Computer Science at Simon Fraser University and join SFU SAT.
- Developed an automated system using Python and Selenium to efficiently send 200 personalized emails to schedule mentoring sessions within a day.
- Engaged students as a computer peer tutor in explorative discussions, encouraging curiosity in introductory programming concepts in Python.
- Collaborated closely with instructors to curate personalized three-hour one-on-one tutoring sessions.

CERTIFICATIONS & AWARDS

- AWS Certified AI Practitioner
- Amateur Radio Operator Certificate
- Computer Science Undergraduate Society Award (one of five awarded annually for SFU contributions)
- Recognition Award from the Mayor of Richmond for Community Contributions
- Global Skills Opportunity: Equity, Diversity, and Inclusion International Mobility Award

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, Bash, HTML, CSS, JSON, XML

Data and APIs: Denodo, Power BI, Pandas, Spark, GraphQL, REST, SOAP

Frameworks and Testing: React, Node.js, Prisma, Selenium, JUnit, Pytest, Cypress

Cloud and DevOps: AWS, Docker, Kubernetes, OpenShift, Jenkins, Bamboo, Bitbucket, IBM WebSphere

Tools: Git, Jira, Postman, SoapUI, Figma, Maven, Poetry